
Detailed Curriculum Vitae

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1. Overall summary

I am an Assistant Professor at the Centre for Quantum Science and Technology (CQST) and the Centre for Security, Theory and Algorithmic Research (CSTAR) at International Institute of Information Technology (IIIT) Hyderabad. My research area contains investigations in the domain of quantum information science with special impetus towards the thermodynamic and statistical analysis of information processing, hierarchical relationships between thermodynamic laws and information theoretic no-go theorems, various quantum resource theories, quantum Markovian and non-markovian dynamics, quantum device engineering, cavity quantum electrodynamics (QED) in connection with quantum devices and information processing etc.

Since joining IIIT Hyderabad in December 2019, I have developed a very active research program in quantum device engineering and thermo-electronics, often in collaboration with faculties and researchers both within and outside of the institute. I have been able to establish a dynamic research collaboration within very senior faculties associated with CSTAR, CQST and CCNSB centres. I have also been able to establish a research program with experiment physicists from IEST, Shibpur, West Bengal. We have already jointly published works in internationally acclaimed journals like Physical Review A and on our way to establish a very fruitful continuing joint theoretical and experimental research program in the domain of cavity QED and quantum opto-mechanics. In my time at IIIT Hyderabad, our newly established CQST centre has been able to assert itself as one of the best research groups in the country. The credit for that is of course shared between all the faculties associated in the group (along with the actively supportive administration of the institute), of which I am a proud active participant. To date, I have published **15** papers in **American Physical Society** journals including **Physical Review Letters**, **Physical Review A** and **Physical Review E**. I also have 10 publications in **IOP science** journals like **Journal of Physics A**, **Journal of Physical Communications**, **Physica Scripta** etc. Overall I have around 40 publications in top rated international physics journals. In my time at IIIT Hyderabad, I have published around 20 papers in top

rated journals like Physical Review Letters, Physical Review A and Journal of Physics A.

At IIIT Hyderabad, I regularly teach both core and advanced elective courses with strong teaching evaluations. From the academic year 2020-21, I have consistently taught three courses per year. Along with that I am supervising (and co-supervising) two PhD students, both of whom will be completing their thesis work in the coming academic year with very good publications. I have also supervised (and co-supervised) the research thesis of 6 Masters / Dual degree students all of whom, with the exception of one (he has submitted his thesis) have already completed their degree. Currently I am supervising two students. Additionally, I am also a co-PI of government research funding from MeitY. I am also a member PI of two National Quantum Mission proposals which are under consideration. I regularly review for international journals like Physical Review Letters, Physical Review A and many others.

2. Personal Statement

My research is concentrated on a few different aspects of quantum information and computational sciences. Primarily I am working in quantum device engineering from the experimental backdrop of cavity QED. The focus of this research program is developing state of the art quantum hardware models and integrated circuits. We closely work with experimentalists in the area of quantum optics, validating our theoretical modelling. Our long term goal is to build a cavity QED based quantum microprocessor. But in the short term, we are concentrating on more fundamentals circuit designings, e.g. quantum operational amplifiers. From the perspective of quantum machine learning and artificial intelligence, we are investigating cavity QED based multi-layer perceptron, quantum reservoir engineering etc. Experimental implementation of such models is one of our primary goals in the coming few years. Secondly, I am also working in the domains of entanglement theory, quantum resource theories, quantum ergodic theory and quantum information processing in general.

Other than concentrating on state of the art quantum research, I also teach core and advanced elective subjects, which are essential to the foundational development of

our students. For the last five years, I have been teaching the UG1 real analysis core course. Additionally I teach an advanced elective course on Linear Partial differential equations and calculus of variance.

I am also involved in UGEE question preparation, reviewing and interviewing applicants for both UG and dual degree. Here at IIIT Hyderabad, in the coming future, I see myself involved in the development of CQST as the leading quantum information and computation group in the country, preferentially in the domain of quantum hardware application

3. Major highlights of my works

Quantum thermo-electric circuit designing: One of our main focus of research is developing quantum thermo-electric integrated circuits. The current state of the art of the research focuses on developing the core components of quantum computing qubits, quantum gates, registers, and chipsets. In the meantime it also emphasizes on the critical role of advanced architectures, such as Field-Programmable Gate Arrays (FPGAs) and Digital Signal Processors (DSPs), in the task of optimizing quantum information processing. Despite their transformative potential, quantum hardware devices (QHD) face critical challenges, such as decoherence, Quantum Error Correction (QEC), scalability, and seamless integration with classical computing systems. We focus on developing the architecture of heat current driven quantum circuits, which mimic the current voltage characteristics of their electronic counterparts. The advantage of such a kind of architecture lies in the fact that it operates in the steady state region and hence robust from the perspectives of decoherence like phenomena. In a recent work with my collaborators from IIT Rajasthan, we have developed a universal approach for the quantum thermal analogue of electric circuits (Phys. Rev. Lett. 135 (2), 020404 (2025)). This work is a stepping stone in the direction of quantum device engineering via thermal manipulation.

Quantum opto-mechanical device engineering: Another very important aspect of our research is closely related to experimental domains like cavity QED. Here we are modelling various quantum devices (both thermally and electronically driven) from the backdrop of cavity atom interactions. It is well known that Quantum optics plays an essential role in the domain of quantum metrology, a fast developing field undergoing exponentially increasing precision in a very short period of time.

Moreover, quantum optics and quantum computation sciences are closely related to one another. Frequently, breakthroughs in quantum optics are implemented in quantum information processing tasks, storage and quantum communication devices. These ongoing developments encouraged us to undergo strongly experimentally motivated quantum device engineering research programs. In this regard, I should mention our recent work, where in collaboration with Subhadip Mitra, CCNSB, IIIT Hyderabad and Rik Chattopadhyay from IEST, Shibpur, Howrah, West Bengal, we developed a non-Markovian model of Quantum Otto cycle, a specific type of quantum heat engine by manipulating the interaction between an atomic system and mirror vibration modes inside a cavity (Phys. Rev. A 112, 042222 (2025)). The atomic system can be a quantum dot, nano-crystals or ultra-cold atoms like rubidium. Further works are in progress, where we are thoroughly developing the thermodynamic processes, e.g. adiabatic, isothermal, isochoric, isobaric etc., in the specific cavity-QED systems. Upon completion, these works will offer an exhaustive description (also possible experimental implementations) of various thermodynamic processes and cycles in cavity-atom based quantum devices.

Quantum dynamics and resource theories: A third more theoretically motivated area of our study is in the domain of various quantum resource theories and quantum dynamics. In this connection, it is important to mention our recent work in exploring non-Markovian ergodic channels and resource theory of activity (Phys. Rev. A 111, 032416 (2025)). Here we have developed a dynamical description of ergodic quantum channels in arbitrary dimensional systems and connected it with a large class of quantum passivity preserving channels. This work not only presents a thorough well defined description of quantum ergodicity, but also furthers the active theoretical research in the domain of the resource theory of active quantum states.

4. Summary of my achievements

State of the art research and international visibility: In my tenure at IIIT Hyderabad, I have co-authored around 20 journal papers in internationally acclaimed and influential journals like **Physical Review Letters**, **Physical Review A**, **Journal of Physics A**, in the domains of quantum thermodynamics and resource theories, Quantum entanglement theory, Quantum device engineering. Overall I have around 40 publications including many publications in top level physics journals. I

have also presented our works in various international conferences attracting visibility, which shows in the current boost of citation of those related articles.

Teaching and supervision: In the course of the last five years, I have supervised the thesis of 6 dual-degree / Masters students who have completed their thesis work. Five of them have completed their degree and one is currently awaiting his thesis defence. Currently I am supervising the thesis of two other students. I am also supervising two PhD students, who will finish their thesis work within the current academic year. I have also regularly taught both core and advanced elective courses in the past five years. I have taught the UG1 real analysis course (core), Linear partial differential equations and calculus of variance (advanced elective) and Open quantum systems and quantum thermodynamics (elective) for the past five years.

Securing external grants: I am currently serving as a co-PI of government research funding from MeitY (Grant No. 4(3)/2024- ITEA). I also have two active proposals in the ongoing National Quantum Mission grant.

Service and institution building: I actively review for leading journals like PRL, PRA, PRB, PRE, Quantum and Quantum Information Processing. In IIIT Hyderabad I contribute in institution building via the responsibilities assigned to me like UGEE question preparation, serving in interview committees for various undergraduate and post graduate programs.

5. Biographical information

Experience:

- **December 2019 to present:** Assistant Professor, CQST and CSTAR, IIIT Hyderabad.
- **August 2019 to December 2019:** Research associate, CeNT, University of Warsaw, Poland.
- **October 2017 to August 2019:** National Post Doctoral Fellow, S.N Bose National Centre for Basic Sciences, Kolkata.
- **April 2015 to October 2017:** Post Doctoral Fellow, Quantum Information and Computation group, Harish Chandra Research Institute, Uttar Pradesh, India.

Education:

- **June 2015:** PhD (in Physical Sciences), University of Calcutta (Host and funding institute: Indian Statistical Institute, Kolkata).
- **August 2008:** MSc. (in Physics), University of Calcutta.
- **August 2006:** BSc. (Hons.) in Physics, Asutosh College, University of Calcutta.

RESEARCH

1. Research statement

With the advent of quantum science and technology in the latter half of the twentieth century came the idea of reversible computation, which paved the way for a new type of computer that is essentially quantum in nature. In such devices, the power of ‘quantum parallelism’ generated by quantum superpositions will lead to new algorithms and complexity classes. The exceptional properties of these devices are illustrated by various quantum algorithms like Shor’s factorization algorithm, which is polynomial in time, in contrast to the factorization algorithms that run on classical Turing computers. These discoveries have significant consequences for the security of many modern cryptographic systems. The fundamental components of quantum computing are then introduced, e.g., reversible logic gates, quantum registers, etc. The key quantum property of ‘entanglement’ can generate tremendous advantages over classical computers. It is imperative to develop hardware that can execute computational tasks that are purely quantum to pave the way for the realization of an actual quantum computer. That is to say, there is a need to develop quantum analogs of the electronic integrated circuits (IC), which are the building blocks of a microprocessor. The central motivation of the current project is to pave the way for a full-fledged quantum device engineering analogous to its electronic counterpart. Keeping these goals in mind, we consider the following research objectives (RO) for the proposed project.

Objectives:

- Constructing the quantum models of basic units of an integrated circuit such as diodes, transistors, rectifiers, LCR circuits, etc.
- Proposing and profiling quantum integrated circuits such as operational amplifiers, oscillators, or quantum analogs of electronic bridges, etc.

- Testing the efficiencies of the proposed devices against the backdrop of various system environment interaction models, like the spin-boson, fermionic bath, and quantum collision models, etc.
- Progressing towards steady-state analysis of complex quantum integrated circuit models via quantum search algorithms, e.g., quantum variational algorithm.

In order to achieve the objectives of our proposed research, we need to construct the dynamics of both electronic and photonic systems trapped in a cavity-like environment, or interacting with spin chains in various different configurations. In order to do that, we will focus on the microscopic derivation of the system dynamics without the Born-Markov approximation and with much less restrictions on the possible energetic transitions and non-static finite dimensional baths. The exact mathematical derivation of such dynamics along with their corresponding master equations will be novel both in their mathematical significance and practical importance. Till date, the dynamical models considered in this particular line of research are more or less phenomenological in nature, because of the lack of presence of exact mathematical procedure. The theoretical aspect of our objectives in this proposal is to bridge this gap significantly.

The theoretical foundation of quantum thermal circuits lies in the arena of open quantum systems. In many-body problems of quantum mechanics, the evolution of microscopic (e.g., spin systems) or mesoscopic (e.g., SQUIDs) systems always gets more involved due to their exposure to a background environment. To realise the reduced dynamics of the quantum system we are interested in, one generally models the environment as a collection of harmonic oscillators or fermionic particles, often defined as a bath. They constitute two different universal classes of quantum environment. In the harmonic-oscillator environment model, the bath is conceived as a set of uncoupled harmonic oscillators. Paradigmatic examples of this kind of bath are the spin-boson and Caldeira-Leggett models, originating from a scheme proposed by Feynman and Vernon. These oscillator models have been widely studied in the context of various physical phenomena under the Markovian approximation. In contrast, the spin-bath models are relatively less explored. However, these models play a pivotal role in the quantum theory of magnetism, quantum spin glasses, the theory of conductors and superconductors, etc. Getting the exact dynamics of a quantum system under the spin-bath model is a highly challenging but interesting (and essential) task. In most cases, the dynamics cannot be described exactly, and several approximate techniques, both local and nonlocal in

time, are used. Other approaches to describe open quantum systems, such as quantum trajectories, the Monte Carlo method, quantum jumps, and the input-output formalism based on stochastic calculus, were also conceived. While these dynamical models are effective and practically realisable, their theoretical expressions depend on complicated mathematical descriptions, often overshadowing the underlying physics. In recent years, however, increasing attention has been paid to much simpler quantum collision models that allow us to focus better on the physical explanations of the model. In the proposed project, we plan to model our quantum heat transport-based device engineering primarily on the following three open quantum systems models.

a. Markovian spin-boson model: In this model, the quantum system interacts with an infinite number of harmonic oscillators, collectively considered the environment. It is one of the most intuitive and common models that can be considered. The model dynamics have been constrained through the imposition of Born-Markov approximation, weak coupling limit, and stationary bath conditions. These restrictions enable us to deal with relatively less complex Markovian Lindblad dynamics, under which the fundamental circuit modeling can be done. We aim to start with such a system-environment interaction, where we can entirely focus on the modelling aspect of the circuits.

Non-Markovian fermionic models: Next, we intend to venture further into more exotic system-bath modelling with fermionic baths available in experimental scenarios that also exhibit non-Markovian traits in their dynamics, making the problem much more intriguing.

Non-Markovian spin-boson model in cavity quantum electrodynamic set up: Here, we will focus on a microscopic derivation of the system dynamics of the spin-boson model, without the Born-Markov approximation and with fewer restrictions on the possible energy transitions, resulting in interesting non-equilibrium situations, which, as we will demonstrate later, can lead to thermo-electric devices offering efficiency enhancement from its Markovian counterpart. In other words, we will generalise the theoretical paradigm beyond the weak coupling and stationary bath approximation to include finite bath models. We will further construct quantum integrated circuits and thermo-electric devices in our setup to demonstrate the possibility of optical cavity-based quantum device engineering with it.

Metrological aspects: One of the intriguing aspects of this study is the possibility of detecting unknown interaction Hamiltonians by mimicking electronic circuits like

the Wheatstone bridge. This allows us to expand the scope of research into quantum metrology of complex systems, encompassing even a part of many body physics beyond the usual domain of quantum information science.

Connection to quantum computation: Quantum systems are always vulnerable to the influence of noisy environments. This influence often causes irreversible dynamics of the system over time, which can be modelled as Markovian master equations. This may lead to a well-defined steady state, irrespective of the initial conditions considered. Simulating various attributes of the stationary state has the same computational complexity as that of isolated quantum systems, which for large systems is not tractable with classical computers. Such problems therefore demand the development of quantum algorithms, suitable to be executed on the next generation of quantum computers. Since we will be dealing with complex open system evolutions needing steady-state analysis for the study of heat current properties, it opens up a scope of marrying two research domains: quantum thermodynamics and quantum computation and algorithms.

2. List of notable 3-4 best publications

- Unravelling the non-Markovian spin-boson model and quantum quasi-Otto cycle, Shreyas Harshal Pradhan, Hadi Mohammed Soufy, Vishal Anand, Rik Chattopadhyay, Subhadip Mitra, **Samyadeb Bhattacharya**, Physical Review A **112** 042222 (2025).
- Quantum Thermal Analogs of Electric Circuits: A Universal Approach, Devvrat Tiwari, **Samyadeb Bhattacharya**, Subhashish Banerjee, Physical Review Letters **135** 020404 (2025).
- Exploring Non-Markovianity in Ergodic Channels: Measuring Memory Retention through Ergotropy, Ritam Basu, Anish Chakraborty, Himanshu Badhani, Mir Alimuddin, **Samyadeb Bhattacharya**, Physical Review A **111** 032416 (2025).
- Emergent non-Markovianity and dynamical quantification of the quantum switch, Vishal Anand, Ananda G. Maity, Subhadip Mitra, **Samyadeb Bhattacharya**, Physical Review A **111** 032428 (2025).

3. Number of Publications:

Type	Number	Annexure
Refereed international journal	39	1
Under submission	0	1
Peer reviewed international conferences	-	-
Books/Reports/White papers/Monographs	-	-
Patents – India	-	-
Patents – International	-	-

Total citations (as of 22.06.2026): 801. H-index: 13, i10-index: 17. More details can be found on my [Google Scholar page](#).

Details of top 5 publications:

- Devrat tiwari, Samyadeb Bhattacharya, **Quantum Thermal Analogs of Electric Circuits: A Universal Approach**, [Phys. Rev. Lett. 135, 020404 \(2025\)](#)

In this Letter, we develop a panoramic schematic of quantum thermal analogs of electric circuits in the steady state regime. We establish the foundations of said premise by defining the analogs of Kirchhoff's laws for heat currents and temperature gradients, as well as a quantum thermal step transformer. Using this, we develop two novel quantum thermal circuits, viz., quantum thermal super Wheatstone bridge and quantum thermal adder circuit, paving the way

for the corresponding integrated circuits. We further show that our approach encompasses various circuits like thermal diode, transistor, and Wheatstone bridge. This sheds new light on the present architecture of quantum device engineering.

- Samyadeb Bhattacharya, Avijit Misra, Chiranjib Mukhopadhyay, Arun Kumar Pati, **Exact master equation for a spin interacting with a spin bath: Non-Markovianity and negative entropy production rate**, [Phys. Rev. A 95, 012122 \(2017\)](#).

An exact canonical master equation of the Lindblad form is derived for a central spin interacting uniformly with a sea of completely unpolarized spins. The Kraus operators for the dynamical map are also derived. The non-Markovianity of the dynamics in terms of the divisibility breaking of the dynamical map and the increase of the trace distance fidelity between quantum states is shown. Moreover, it is observed that the irreversible entropy production rate is always negative (for a fixed initial state) whenever the dynamics exhibits non-Markovian behavior. In continuation with the study of witnessing non-Markovianity, it is shown that the positive rate of change of the purity of the central qubit is a faithful indicator of the non-Markovian information backflow. Given the experimental feasibility of measuring the purity of a quantum state, a possibility of experimental demonstration of non-Markovianity and the negative irreversible entropy production rate is addressed. This gives the present work considerable practical importance for detecting the non-Markovianity and the negative irreversible entropy production rate.

- Avijit Misra, Uttam Singh, Samyadeb Bhattacharya, Arun Kumar Pati, **Energy cost of creating quantum coherence**, [Phys. Rev. A 93, 052335 \(2016\)](#)

We consider physical situations where the resource theories of coherence and thermodynamics play competing roles. In particular, we study the creation of quantum coherence using unitary operations with limited thermodynamic resources. We find the maximal coherence that can be created under unitary operations starting from a thermal state and find explicitly the unitary transformation that creates the maximal coherence. Since coherence is created by unitary operations starting from a thermal state, it requires some amount of energy. This motivates us to explore the

trade-off between the amount of coherence that can be created and the energy cost of the unitary process. We also find the maximal achievable coherence under the constraint on the available energy. Additionally, we compare the maximal coherence and the maximal total correlation that can be created under unitary transformations with the same available energy at our disposal. We find that when maximal coherence is created with limited energy, the total correlation created in the process is upper bounded by the maximal coherence, and vice versa. For two-qubit systems we show that no unitary transformation exists that creates the maximal coherence and maximal total correlation simultaneously with a limited energy cost.

- Titas Chanda, Samyadeb Bhattacharya, **Delineating incoherent non-Markovian dynamics using quantum coherence**, [Annals of Physics 366, 1-12 \(2016\)](#)

We introduce a method of characterization of non-Markovianity using coherence of a system interacting with the environment. We show that under the allowed incoherent operations, monotonicity of a valid coherence measure is affected due to non-Markovian features of the system–environment evolution. We also define a measure to quantify non-Markovianity of the underlying dynamics based on the non-monotonic behavior of the coherence measure. We investigate our proposed non-Markovianity marker in the behavior of dephasing and dissipative dynamics for one and two qubit cases. We also show that our proposed measure captures the back-flow of information from the environment to the system and is compatible with well known distinguishability criteria of non-Markovianity.

- Samyadeb Bhattacharya, Bihalan Bhattacharya, Archan S Majumdar, **Convex resource theory of non-Markovianity**, [Journal of Physics A: Mathematical and Theoretical 54 \(3\), 035302\(2020\)](#)

We establish a convex resource theory of non-Markovianity inducing information backflow under the constraint of small time intervals within the temporal evolution. We identify the free operations and a generalized bona-fide measure of non-Markovian information backflow. The framework satisfies the basic properties of a consistent resource theory. The proposed resource quantifier is lower bounded by the optimization-free

Rivas–Huelga–Plenio (RHP) measure of non-Markovianity. We next define the robustness of non-Markovianity and show that it can be directly linked with the RHP measure of non-Markovianity through a lower bound. This enables a physical interpretation of the RHP measure. We further relate robustness of non-Markovianity with the quantum capacity of dephasing channels.

4. Thesis and FYP of students

Program	Supervising	Supervised	Annexure
PhD	2	0	3
MS by research	1	4	3
Dual degree	1	2	3

I provide below a brief overview of the students I have supervised or am currently supervising. Further details are given in [Annexure 3](#).

PhD: I am primarily supervising two PhD students. **Vishal Anand** is my first PhD student. He is in his 4th year. I am also supervising **Abhishek Muhuri**, who is co-supervised by [Prof. Aditi Sen De](#) from Harish Chandra Research Institute.

MS by research / Dual degree: I have supervising / supervised 8 MS and Dual degree students so far.

- **Graduated:**

1. [Sravani Yanamandra](#), MS by research.
2. [P V Srinidhi](#), MS by research
3. [Ritam Basu](#) MS by research
4. [Vaibhav Chimalgi](#), MS by research
5. [Ankit Vaishy](#), Dual degree (CND).
6. [Shreyas H. Pradhan](#), Dual degree (CND).

- **Currently Pursuing:** Vanshika Alawat, Dual degree (CND), Mokhsata Seth MS by research.

5. Major research projects

Since joining IIIT Hyderabad I have applied for a few external funded projects. Some of which are still under consideration. Other than that I have one successful external project of which I am a co-PI from IIIT Hyderabad.

External grants from the Government:

Name of the project	Type	Funding agency	Sanctioned amount	From	To	Role
Quantum Machine Learning: Use cases and Applications	Collaborative proposal with CDACs (Hyderabad, Pune, Mohali, Silchar, Noida, Delhi) and IITs (Roorkee, Gandhinagar, Ropar)	MeitY SERB (Now ANRF)	93.6 L	2024	2027	co-PI

Internal grants:

Name of the project	Funding agency	Sanctioned amount	From	To	Role
Faculty Seed Fund	IIIT Hyderabad	5L	2020	2023	PI

Honors and awards related to research

Name of award	Awarding agency	Year
Research Associateship	CeNT, University of Warsaw	2019
D S Kothari Post Doctoral Fellowship	University Grant Commission, Govt. of India	2019
National Post Doctoral Fellowship	SERB (now ANRF), Govt. of India	2017
Institute post doctoral fellowship	Harish Chandra Research Institute, Uttar Pradesh, India	2015
Junior Research Fellowship	Indian Statistical Institute	2009
CSIR-NET (LS)	CSIR	2009

TEACHING

1. Teaching statement

In my understanding, teaching is an equally important part of the job for an academic. Research and teaching are two essential aspects of the academic journey of a researcher, always engaging in a symbiotic relationship. As per the responsibility of a faculty of any educational institution, it however enjoys primary importance. Engaging with young minds full of fresh ideas is not only our primary duty, but also a task full of immense pleasure and often pleasant surprises. Moreover, from the standpoint of an active researcher, teaching helps us to experience different perspectives of the same topic, which often helps clarify and strengthen conceptual fundamentals of a subject. Keeping these in mind, I often try to engage with students through basic questions about the subject at hand, sometimes working out instant quizzes, instant problem solving and so on. These types of activities, in my opinion, help develop conceptual understanding and also establish a healthy student-teacher relationship.

New Courses developed:

- I. Open quantum systems and quantum thermodynamics (OQS)
- II. Linear partial differential equations and calculus of variance (LPDE)

2. Courses taught

S. No.	Semester	Course	Course code and level	No of Students	Review (out of 5)	Instructor Review (out of 5)	Co-teaching info
1.	M20	OQS and Quantum	SC1.310	40	NA	NA	-

		thermo- dynamics					
2.	S21	LPDE and calculus of variance	MA4.30 3	24	NA	NA	
3.	M21	OQS and Quantum thermodyna mics	SC1.310	50	NA	NA	
4.	S22	LPDE and calculus of variance	MA4.30 3	32	NA	NA	
5.	M22	OQS and Quantum thermodyna mics	SC1.310 And SC1.410	36	3.9	4	
6.	M22	Real Analysis	MA4.10 1	232	NA	NA	
7.	S23	LPDE and calculus of variance	MA4.30 3	47	3.1	3.4	
8.	M23	OQS and Quantum thermodyna mics	SC1.310 And SC1.410	22	3.78	4	
9.	M23	Real Analysis	MA4.10 1	306	-	-	
10.	S24	LPDE and calculus of variance	MA4.30 3	57	4.3	4.1	
11.	M24	OQS and Quantum thermodyna mics	SC1.310 And SC1.410	11	4.26	5	

12.	M24	Real Analysis	MA4.10 1	318	3.29	3	With Abhishek Sribastava
13.	S25	LPDE and calculus of variance	MA4.30 3	34	4	5	
14.	M25	Real Analysis	MA4.10 1	354	3.48	3.6	With Lalitha Vadlamani and Uttam Singh
15.	S26	LPDE and calculus of variance	MA4.30 3	47	-	-	

3. Teaching goals, method and outcome

When designing and teaching courses at modern information technology driven institutions like IIIT Hyderabad, the faculties who themselves have been taught in more of a traditionally methodical approach have to keep a few important things in mind. Primarily, the bulk of the students (especially undergraduates) who come to study here possess very specific bird eye vision approaches towards their goals and motivations for coming to this place. These groups of students have a mindset purely driven by the motivation of acquiring. On the other hand, there are also a smaller group of students who are motivated towards research oriented learning. As faculties, we have to address the needs of both groups of students by keeping a balance in designing and teaching the curriculum. My primary motivation behind

teaching any courses is of course to convey maximum coherent information about the subject to the students. But keeping the need of balancing the curriculum in mind, I employ some specific strategies when teaching. Firstly, I always present the subjects to the students with pin pointed objectives to be achieved at the end of the course. If I am teaching a mathematical topic, I always present it with its applications, so that a student always has a rational understanding of why he/she is supposed to learn certain mathematical methods. I always try to convey the message that in what specific situation in their problem solving in professional domains, the methods might be found to be handy. This is what I always do, when teaching courses like Real analysis or Linear partial differential equations (LPDE). Apart from that I teach another advanced elective course on open quantum systems (OQS), which is closer to my own research domains. This particular course is much more research oriented and designed to target students who are more academically motivated and want to pursue research. In teaching this course, parallelly with the standard curriculum, I also try to impress upon the fundamentals of research methodology.

All my courses are divided into a few evaluation parameters. The standard two quizzes, mid-semester and end-semester exams are the main evaluation parameters. Along with that I also give 2-3 assignments per semester. In seating exams I usually divide the questions in two specific patterns. 60% to 70% of exam questions are standard, done in class type of problems and 30% to 40% of the questions are more intricately designed which requires a little bit of out of the book thinking. This is a fair way of giving incentives and hence encouraging the above average students, but also not letting the rest of the students to suffer severely. In the more research oriented courses, I also hand out reading projects concerning unravelling some basic research articles and delivering an end semester presentation on that, as a course project.

The outcome of the courses can be readily understood from the student feedback of the courses throughout my tenure in IIIT Hyderabad. The feedbacks for my courses has consistently been high, as reflected in the previous table, for the total period of my teaching here.

4. Developing new courses and curriculum

After coming to IIIT Hyderabad, I have designed two elective courses, previously not offered here. First is my course on Open Quantum Systems (OQS) and quantum thermodynamics, which is a research oriented course, primarily targeted towards CQST and CCNSB students, who are interested in quantum science and technologies, though not at all confined with these specific groups of students. The curriculum for this course is designed in such a way that mathematically or Physics wise less inclined students can also take the course and do well (and they have done so consistently). The second course that I designed is a maths elective course on Linear partial differential equations (LPDE) and calculus of variance.

Going forward, I intend to offer more courses both research oriented and also more leaning towards engineering applications. For example, I am planning to propose a course in the theory and application of cavity quantum electrodynamics from the perspective of quantum technologies. This course will be an upgraded version of my current OQS course, which will prepare students for current state of the art technologies for quantum information and computation applications. Moreover, I am also planning to upgrade my LPDE course with applications in bio-physics, environmental ecology etc.

5. Honors and awards related to teaching

I have yet to receive any formal awards related to teaching, from the institute or otherwise, but the feedback, intense interactions and appreciations from my students is reward enough for me. It perpetually encourages me to improve upon my skill of teaching and critically analyzing my methods of teaching.

SERVICE

1. Service statement

I consider service both in the capacity of an administrator and other capacities, as an integral part of an academic career, alongside teaching and research responsibilities. It is my conviction that I have fulfilled all the responsibilities bestowed upon me by the institution administration alongside any other obligations as an academic. Within IIIT Hyderabad, I have been assigned responsibilities like preparing and reviewing UGEE entrance examination questions, serving as panel member of UG/PG entrance interviews, for the past six years. More broadly, I volunteer for academic services like reviewing for many top tier international journals, and also bear responsibilities of serving as committee members of several critically acclaimed national conferences.

2. Professional and societal services

❖ Organising conferences

Name of the conference	Level	Year	Location	Role
National Quantum Science and Technology Symposium (NQSTS)	National	2021	IIIT Hyderabad	Organising committee
Quantum Enabled Science and	International	2022	IIIT Hyderabad	Local organizing

Technology Symposium (QUEST)				committee
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❖ Paper reviewing

Journal

Name of the journal	Impact factor
Physical Review Letters	9.0
Physical Review A	2.9
Physical Review B	3.7
Physical Review E	2.4
Journal of Physics A(Mathematical and Theoretical)	2.0
Journal of Physics B	1.798
Quantum	6.777
Quantum Information Processing journal	2.2

❖ Delivering lectures

I have delivered numerous lectures in different international and national conferences throughout my academic career. Following is a brief list of them.

Theme of the lecture	Type of talk	Location	Event
Unraveling the non-Markovian spin-boson model and quantum quasi-Otto cycle	Invited	IISER Mohali	Quantum Foundations Technology and Applications 2025
Emergent non-Markovianity and dynamical quantification of the quantum switch	Invited	HRI Uttar Pradesh	Quantum Information Processing and Applications 2023

3. IIIT services

In my tenure at IIIT Hyderabad, I have given my service as assigned to me by the administration, to the best of my abilities. From the academic year of 2020-21, I have consistently served as a panel member in various interview boards including UG, lateral entry, PhD entrance etc. I am also consistently involved in preparation and review of entrance examination question papers.

Annexure 1:

List of publications and pre-prints

Published during the tenure at IIIT Hyderabad

- Unravelling the non-Markovian spin-boson model and quantum quasi-Otto cycle, Shreyas Harshal Pradhan, Hadi Mohammed Soufy, Vishal Anand, Rik Chattopadhyay, Subhadip Mitra, **Samyadeb Bhattacharya**, [Phys. Rev. A 112, 042222 \(2025\)](#)
- Quantum Thermal Analogs of Electric Circuits: A Universal Approach, Devvrat Tiwari, **Samyadeb Bhattacharya**, Subhashish Banerjee, [Phys. Rev. Lett. 135 \(2\), 020404 \(2025\)](#)
- Exploring Non-Markovianity in Ergodic Channels: Measuring Memory Retention through Ergotropy, Ritam Basu, Anish Chakraborty, Himanshu Badhani, Mir Alimuddin, **Samyadeb Bhattacharya**, [Phys. Rev. A 111, 032416 \(2025\)](#)
- Using Quantum Switches to Mitigate Noise in Grover's Search Algorithm, Suryansh Srivastava, Arun K. Pati, **Samyadeb Bhattacharya**, Indranil Chakrabarty, [J. Phys. A: Math. Theor. 58 105304 \(2025\)](#)

- Breaking absolute separability with quantum switch, Sravani Yanamandra, P V Srinidhi, **Samyadeb Bhattacharya**, Indranil Chakrabarty, Suchetana Goswami, [Quantum Inf Process 24, 81 \(2025\)](#)
- Emergent non-Markovianity and dynamical quantification of the quantum switch, Vishal Anand, Ananda G. Maity, Subhadip Mitra, **Samyadeb Bhattacharya**, [Phys. Rev. A 111, 032428 \(2025\)](#)
- Interplay between the Hilbert-space dimension of a control system and the memory induced by a quantum SWITCH, Saheli Mukherjee, Bivas Mallick, Sravani Yanamandra, **Samyadeb Bhattacharya**, Ananda G. Maity, [Phys Rev A.110.042624 \(2024\)](#)
- Quantum channels that destroy negative conditional entropy, PV Srinidhi, Indranil Chakrabarty, **Samyadeb Bhattacharya**, Nirman Ganguly, [Phys. Rev. A 110 \(2024\), 042423](#)
- Relating CP divisibility of dynamical maps with compatibility of channels, Arindam Mitra, Debashis Saha, **Samyadeb Bhattacharya**, A. S. Majumdar, [Phys. Rev. A 109, 062213 \(2024\)](#)
- Activating information backflow with the assistance of quantum SWITCH, Ananda G. Maity, **Samyadeb Bhattacharya** , [J. Phys. A: Math. Theor. 57 215302 \(2024\)](#)
- Detecting entanglement harnessing Lindblad structure, Vaibhav Chimalgi, Bihalan Bhattacharya, Suchetana Goswami, **Samyadeb Bhattacharya** , [Phys. Scr. 98 115117 \(2023\)](#)

- Non-Markovianity and entanglement detection, Sourav Chanduka, Bihalan Bhattacharya, Rounak Mundra, **Samyadeb Bhattacharya**, Indranil Chakrabarty, [International Journal of Quantum Information](#) Vol. 21, No. 05, 2350025 (2023)
- Dynamics of two central spins immersed in spin baths, Devvrat Tiwari, Shounak Datta, **Samyadeb Bhattacharya**, Subhashish Banerjee, [Phys. Rev. A](#) 106, 032435 (2022)
- Detecting genuine multipartite entanglement in three-qubit systems with eternal non-Markovianity, Ankit Vaishy, Subhadip Mitra, **Samyadeb Bhattacharya**, [J. Phys. A: Math. Theor.](#) 55 225305 (2022)
- Generating and detecting bound entanglement in two-qutrits using a family of indecomposable positive maps, Bihalan Bhattacharya, Suchetana Goswami, Rounak Mundra, Nirman Ganguly, Indranil Chakrabarty, **Samyadeb Bhattacharya**, A. S. Majumdar, [J. Phys. Commun.](#) 5 065008 (2021)
- Nearly Markovian maps and entanglement-based bound on corresponding non-Markovianity, Sreetama Das, Sudipto Singha Roy, **Samyadeb Bhattacharya**, Ujjwal Sen, [J. Phys. A: Math. Theor.](#) 54 395301 (2021)
- Thermodynamic utility of Non-Markovianity from the perspective of resource interconversion, **Samyadeb Bhattacharya**, Bihalan Bhattacharya, A. S. Majumdar, [J. Phys. A: Math. Theor.](#) 53 335301 (2021)
- Convex geometry of Markovian Lindblad dynamics and witnessing non-Markovianity, Bihalan Bhattacharya, **Samyadeb Bhattacharya**, [Quantum Information Processing](#), 20 253 (2021)

- Detecting non-Markovianity via uncertainty relations, Ananda G. Maity, **Samyadeb Bhattacharya**, A. S. Majumdar, [Journal of Physics A: Mathematical and Theoretical \(2020\)](#)
- Convex resource theory of non-Markovianity, **Samyadeb Bhattacharya**, Bihalan Bhattacharya, A S Majumdar, [J. Phys. A: Math. Theor. 54 035302 \(2020\)](#)
- Wave-particle duality employing quantum coherence in superposition with non-orthogonal pointers, Sreetama Das, Chiranjib Mukhopadhyay, Sudipto Singha Roy, **Samyadeb Bhattacharya**, Aditi Sen De, Ujjwal Sen, [J. Phys. A: Math. Theor. 53 115301\(2020\)](#)

Published before the tenure at IIIT Hyderabad

- Quantum speed limit constraints on a nanoscale autonomous refrigerator, Chiranjib Mukhopadhyay, Avijit Misra, **Samyadeb Bhattacharya**, Arun Kumar Pati, [Phys. Rev. E 97, 062116 \(2018\)](#)
- Universal quantum uncertainty relations between non-ergodicity and loss of information, Natasha Awasthi, **Samyadeb Bhattacharya**, Aditi Sen De, Ujjwal Sen, [Phys. Rev. A 97, 032103 \(2018\)](#)
- Dynamics and thermodynamics of a central spin immersed in a spin bath, Chiranjib Mukhopadhyay, **Samyadeb Bhattacharya**, Avijit Misra, Arun Kumar Pati, [Phys. Rev. A 96, 052125 \(2017\)](#)

- Exact master equation for a spin interacting with a spin bath: Non-Markovianity and negative entropy production rate, **Samyadeb Bhattacharya**, Avijit Misra, Chiranjib Mukhopadhyay, Arun Kumar Pati, [Phys. Rev. A 95, 012122 \(2017\)](#)
- Energy Cost of Creating Quantum Coherence, Avijit Misra, Uttam Singh, **Samyadeb Bhattacharya**, Arun Kumar Pati, [Phys. Rev. A 93, 052335 \(2016\)](#)
- Evolution of coherence and non-classicality under global environmental interaction, **Samyadeb Bhattacharya**, Subhashish Banerjee, Arun Kumar Pati, [Quantum Inf Process \(2018\) 17: 236](#)
- Delineating incoherent non-Markovian dynamics using quantum coherence, Titas Chanda, **Samyadeb Bhattacharya**, [Annals of Physics 366, 1-12 \(2016\)](#)
- Quantum Path predictability for an electronic Mach-Zehnder interferometer in presence of environment induced Decoherence and Quantum Erasing process, **Samyadeb Bhattacharya**, Sisir Roy, [Journal of Modern Physics 7 \(2016\)](#)
- Wigner-Yanase skew information and entanglement generation in quantum measurement, Manik Banik, Prasenjit Deb, **Samyadeb Bhattacharya**, [Quantum Information processing 16 1-12 \(2017\)](#)

- Temperature dependent Zeno time for a two level atom tunneling through a thermal magnetic barrier in the framework of weak measurement, **Samyadeb Bhattacharya**, Sisir Roy, [Journal of Modern Physics .6 9 \(2015\)](#)
- Comparison of decoherence and Zeno dynamics from the context of weak measurement for a two level atom tunneling through squeezed vacuum, **Samyadeb Bhattacharya**, [International Journal of Modern Physics B, Volume 30, page 1650016 \(2016\)](#)
- Preservation of coherence for a Two-Level atom tunneling through a squeezed vacuum with finite bandwidth, **Samyadeb Bhattacharya**, Sisir roy, [Annals of Physics, volume 348, page 15 \(2014\)](#)
- Temperature dependent weak value of dwell time for a two state particle tunneling through a thermal magnetic barrier, **Samyadeb Bhattacharya** , [Phys. Rev. A 89, 022110 \(2014\)](#)
- Laser trapping of ions and asymptotic minimization of Decoherence, **Samyadeb Bhattacharya**, Sisir Roy, [Modern Physics Letters B, Vol. 27, No. 18 \(2013\) 1350137](#)

- Hartman effect and dissipative quantum systems, **Samyadeb Bhattacharya**, Sisir Roy, [J. Math. Phys. 54, 052101 \(2013\)](#)
- Weak value of Dwell time for Quantum Dissipative spin-1/2 System, **Samyadeb Bhattacharya**, Sisir Roy, [Physical Review A 85, 062119 \(2012\)](#)
- Schrödinger-Langevin Equation and Ion Transport at Nano Scale, **Samyadeb Bhattacharya**, Sisir Roy, [Journal of Modern Physics, 2011, 2, 231-235](#)
- Dissipative Effect and Tunneling Time, **Samyadeb Bhattacharya**, Sisir Roy, [Advances in Mathematical Physics 2011 1 \(2011\)](#)

Unpublished works:

- Weak measurement and Zeno time in a dissipative environment, **Samyadeb Bhattacharya**, Sisir Roy, [arXiv1209.0279](#)
- Quantum Thermodynamics and the coherence in Ion channels, **Samyadeb Bhattacharya**, Sisir Roy, [arXiv1303.1905](#)

Annexure 2

List of invited talks, workshops and conferences attended

- Quantum Foundations Technology and Applications 2025, IISER Mohali, India (Invited).
- Quantum Information Processing and Applications 2023, HRI, Uttar Pradesh, India (Invited).
- Quantum Enabled Science and Technology Symposium ([QUEST](#)) 2022, IIIT Hyderabad, India (attended, host).
- National Quantum Science and Technology Symposium ([NQSTS](#)) 2021, IIIT Hyderabad, India (attended, host).
- Meeting on Quantum information processing and applications (QIPA) 2018, HRI, Uttar Pradesh, India (Invited).
- International symposium on new frontiers in quantum correlations 2018, SNBNCBS, Kolkata, India (presented).
- Advanced school and workshop on Quantum Science and Quantum technologies 2017, ICTP, Italy (presented).
- Physics and Applied Mathematics Research scholars meet 2017, ISI, Kolkata, India (Invited).
- Recent trends in quantum theory 2017, University of Calcutta, India (Presented).
- Young Quantum 2017, HRI, Uttar Pradesh, India (presented).
- Physics and Applied Mathematics Research scholars meet 2015, ISI, Kolkata, India (Invited).
- Young Quantum 2015, HRI, Uttar Pradesh, India (Invited).

- Asian Quantum Information Science conference 2013, Chennai, India (poster presentation).
- Frontiers in Quantum and Mesoscopic Thermodynamics 2013, Prague (poster presentation).

Annexure 3

Student Supervision

I am currently supervising two PhD students, who are expected to graduate in 2026.

Vishal Anand (December 2021- present)

Program: Computational Natural Sciences (CNS), IIIT Hyderabad

Thesis title: An exploration on thermodynamic aspects of quantum processes.

Qualification: Integrated MSc in Mathematics and computing, BIT MISRA, Ranchi, Jharkhand.

Abhishek Muhuri (2022-present)

Program: Computational Natural Sciences (CNS), IIIT Hyderabad

Thesis title: Not decided yet.

.Qualification: MSc in Physics IIISER Kolkata, Mohonpur , West Bengal.

MS by research / Dual degree thesis

As of now I have supervised (co-supervised) the thesis of six dual degree / MS by research students, among whom five have already successfully defended their thesis. One is due to defend on 28th February, 2026. Currently I am actively supervising (co-supervising) two students.

- **Graduated**

- ❖ Sravani Yanamandra, MS by research.
- ❖ P V Srinidhi, MS by research (co-supervised with I Chakraborty)
- ❖ Ritam Basu, MS by research
- ❖ Vaibhav Chimalgi, MS by research
- ❖ Ankit Vaishy, Dual degree (co-supervise with S. Mitra)
- ❖ Shreyas H Pradhan, Dual degree (thesis defence on February 28, 2026)

- **Currently pursuing**

- ❖ Vanshika Alawat, Dual degree (co-supervised with S. Mitra)
- ❖ Mokshda Seth, MS by research

Major research Project

Quantum Machine Learning: Use-cases and Applications

- **Project duration:** 3 years (2024 - 2027)
- **Status:** Ongoing
- **Funding Agency:** Ministry of Electronics and Information Technology (MeitY), Government of India.
- **Institutional partners:** IIT Roorkee, IIT Gandhinagar, IIT Ropar, CDAC (Hyderabad, Mohali, Silchar, Noida, Delhi)
- **Sanctioned Budget for IIIT:** 93.6 L (Total sanctioned budget for all members: 17.8 Cr)
- **Role:** I am one of the co-PIs from IIIT Hyderabad.
- **Other Members from IIIT:** The following CQST members are Co-PI: Uttam Singh, Siddhartha Das, Indranil Chakrabarty, Subhadip Mitra, Harjinder Singh, Kannan Srinathan. Shantanav Chakraborty is leading the project from IIIT Hyderabad as the PI.

- **Objectives for IIIT:** In the proposal, each PI has their own research objectives to fulfill. I am stating therefore, my own research objectives in the following.

- ❖ Constructing the quantum models of basic units of an integrated circuit such as diodes, transistors, rectifiers, LCR circuits, etc.
- ❖ Proposing and profiling quantum integrated circuits such as operational amplifiers, oscillators, or quantum analogs of electronic bridges, etc.
- ❖ Testing the efficiencies of the proposed devices against the backdrop of various system environment interaction models, like the spin-boson, fermionic bath, and quantum collision models, etc.
- ❖ Progressing towards steady-state analysis of complex quantum integrated circuit models via quantum search algorithms, e.g., quantum variational algorithm.

- **Major highlights:**

- ❖ **Publications:** Multiple publications in top venues from the IIIT team such as Physical Review Letters, Quantum, PRX Quantum, Quantum, Quantum Science and Technology.
- ❖ **High impact research:** Under the IIIT proposal, we have developed a panoramic schematic of quantum thermal analogs of electric circuits in the steady state regime. We establish the foundations of said premise by defining the analogs of Kirchhoff's laws for heat currents and temperature gradients, as well as a quantum thermal step transformer. Using this, we develop two novel quantum thermal circuits, viz., quantum thermal super Wheatstone bridge and quantum thermal adder circuit, paving the way for the corresponding integrated circuits. Moreover, we have constructed an experimentally feasible cavity-QED

based model of quantum Otto engine, paving the way to develop a comprehensive cavity-QED based quantum device engineering mechanism.

Work experience(in chronological order).

S.No	Position Held	Name of the Institute	From	To	Payscale (gross salary /month)
1.	JRF	Indian Statistical Institute	2009	2010	Rs.15600
2.	JRF	Indian Statistical Institute	2010	2011	Rs.20800
3.	SRF	Indian Statistical Institute	2011	2015(March)	Rs23400
4.	PostDoctoral Fellow	Harish Chandra Research Institute	2015(April)	2016 (March)	Rs36000
5.	PostDoctoral Fellow	Harish Chandra Research Institute	2016(April)	2017(March)	Rs38000
6.	PostDoctoral Fellow	Harish Chandra Research Institute	2017(April)	2017 (September)	Rs40000
7.	NationalPost Doctoral Fellow (SERB)	S.N.B.N.C.B. S.	2017 (October)	2019 (August)	Rs55000
8.	Research Associate	CeNT, University of Warsaw	2019 (August)	2019 (November)	8500 PLN (Polish currency)

9.	Assistant Professor	IIIT Hyderabad	2019 (December)	Till Date	Rs.250000 (including taxes, approx. and changing as per rule)
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List of major publications (in chronological order).

S.No	Author(s)	title	Name Of Journal	Volume	Page	Year
1.	Shreyas Harshal Pradhan, Hadi Mohammed Soufy, Vishal Anand, Rik Chattopadhyay, Subhadip Mitra, Samyadeb Bhattacharya	Unravelling the non-Markovian spin-boson model and quantum quasi-Otto cycle	Physical Review A	112	042222	2025
2.	D Tiwari, Samyadeb Bhattacharya, Subhashish Banerjee	Quantum thermal analogs of electric circuits: A universal approach	Physical Review Letters	135	020404	2025
3.	R Basu, A Chakraborty, H Badhani, M Alimuddin, Samyadeb Bhattacharya	Exploring Non-Markovianity in Ergodic Channels: Measuring Memory Retention through Ergotropy	Physical Review A	111	032416	2025

4.	S Yanamandra, P V Srinidhi, Samyadeb Bhattacharya, I chakraborty and S Goswami	Breaking absolute separability with quantum switch	Quantum Information Processing	24	81	2025
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5.	V Anand, A G Maity, S Mitra and Samyadeb Bhattacharya	Emergent non-Markovianity and dynamical quantification of the quantum switch	Physical Review A	111	032428	2025
6.	S Srivastava, A K Pati, Samyadeb Bhattacharya and I Chakraborty	Using Quantum Switches to Mitigate Noise in Grover's Search Algorithm	Journal of Physics A: Mathematical and Theoretical	58	105304	2025
7.	S Mukherjee, B Mallick, S Yanamandra, Samyadeb Bhattacharya and AG Maity	Interplay between the Hilbert-space dimension of a control system and the memory induced by a quantum SWITCH	Physical Review A	110	042624	2024
8.	P V Srinidhi, I Chakraborty, Samyadeb Bhattacharya and N Ganguly	Quantum channels that destroy negative conditional entropy	Physical Review A	110	042423	2024
9.	A Mitra, D Saha, Samyadeb Bhattacharya and A S Majumdar	Relating CP divisibility of dynamical maps with compatibility of channels	Physical Review A	109	062213	2024
10.	A G Maity and Samyadeb Bhattacharya	Activating information backflow with the assistance of quantum SWITCH	Journal of Physics A: Mathematical and Theoretical	57	215302	2024
11.	S Chanduka, B Bhattacharya, R Mundra, Samyadeb Bhattacharya and I Chakraborty	Non Markovianity and entanglement detection	International Journal of Quantum Information	21	2350025	2023

12.	Vaibhav Chimalgi, Bihalan Bhattacharya ,Suchetana Goswami, Samyadeb Bhattacharya	Detecting entanglement harnessing Lindblad structure	Physica Scripta	98	115117	2023
13.	Devvrat Tiwari, Shounak Datta, Samyadeb Bhattacharya ,Subhashish Banerjee	Dynamics Of two central spins immersed in spin baths	Physical Review A	106	032435	2022
14.	Ankit Vaishy, Subhadip Mitra, Samyadeb Bhattacharya	Detecting genuine multipartite entanglement in three-qubit systems with external non-Markovianity	Journal Of Physics A	55	225305	2022
15.	Bihalan Bhattacharya, Suchetana Goswami, Rounak Mundra, Nirman Ganguly, Indranil Chakrabarty, Samyadeb Bhattacharya, A.S. Majumdar	Generating and detecting entanglement in two-qutrits using family indecomposable positive maps	Journal of Physics communication	5	065008	2021
16.	S Das,SS Roy, Samyadeb Bhattacharya ,and U Sen	Nearly Markovian Maps and entanglement-based bound on corresponding non-Markovianity	Journal Of Physics A	54	395301	2021
17.	Bihalan Bhattacharya and Samyadeb Bhattacharya	Convex geometry of Markovian Lindblad dynamics and witnessing non-Markovianity	Quantum Information Processing	20	253	2021

18.	Samyadeb Bhattacharya ,Bihalan Bhattacharya and AS Majumder	Thermodyna mic utility of Non-Markovi anity From The Perspective Of resource interconversi on	Journal Of Physics A	53	335301	2020
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19.	AG Maity, Samyadeb Bhattacharya and AS Majumder	Detecting non-Markovia nity via uncertainty relations	Journal Of Physics A	53	175301	2020
20.	Samyadeb Bhattacharya ,Bihalan Bhattacharya and AS Majumder	Convex resource theory of non-Markovi anity	Journal Physics A	54	035302	2020
21	S Das, C Mukhopadhya y, SS Roy, Samyadeb Bhattacharya , A Sen De and U Sen	Wave-particle duality employing quantum coherence in superposition with non-orthogon al pointers	Journal Of Physics A	53	115301	2020
22	C Mukhopadhya y, A Misra, Samyadeb Bhattacharya and AK Pati	Quantum speed limit constraints on a nanoscale autonomous refrigerator	Physical Review E	97	062116	2018
23	N Awasthi, Samyadeb Bhattacharya ,A Se nDe and U Sen	Universal quantum uncertainty relations between non-ergodici t y and loss of information	Physical Review A	97	032103	2018

24	Samyadeb Bhattacharya ,S Banerjee and AK Pati	Evolution Of Coherence And non-classicali ty under global environmenta l interaction	Quantum Information Processing	17	236	2018
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25	C Mukhopadhy ay, Samyadeb Bhattacharya , A Misra,and AK Pati	Dynamics And Thermodyna mics of a central spin immersed in a spin bath	Physical Review A	96	052125	2017
26	Samyadeb Bhattacharya ,A Misra, C Mukhopadhya y and AK Pati	Exact master equation for a spin interacting with spa bath: Non-Markovi anity and negative entropy production rate	Physical Review A	95	012122	2017

27	M Banik, P.Deb and Samyadeb Bhattacharya	Wigner-Yana se skew information and entanglement generation in quantum measurement	Quantum Information Processing	16	97	2017
28	A Misra, USingh, Samyadeb Bhattacharya and A KPati	Energy Cost of creating quantum coherence	Physical Review A	93	052335	2016
29	T Chanda and Samyadeb Bhattacharya	Delineating incoherent non-Markovian dynamics using quantum coherence	Annals of Physics	366	1	2016

30	Samyadeb Bhattacharya	Comparison of decoherence and Zeno dynamics from the context of weak measurementf or a two Level atom tunneling through squeezed vacuum	International Journal of Modern Physics B	30	1650016	2016
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31	Samyadeb Bhattacharya and S.Roy	Preservation Of Coherence for a Two-Level Atom Tunneling Through a squeezed vacuum with finite bandwidth	Annals of Physics	348	15	2014
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32	Samyadeb Bhattacharya	Temperature dependent weak value of dwell time for a two state particle tunneling through a thermal magnetic barrier	Physical ReviewA	89	022110	2014
33	Samyadeb Bhattacharya and S.Roy	Laser Trapping of ions and asymptotic minimization Decoherence	Journal of Modern Physics B	27	1350137	2013
34	Samyadeb Bhattacharya and S.Roy	Hartman Effect and dissipative quantum systems	Journal Of Mathematical Physics	54	052101	2013
35	Samyadeb Bhattacharya and S.Roy	Weak value of Dwell time for Quantum Dissipative spin- $\frac{1}{2}$ System	Physical ReviewA	85	062119	2012

